SC7 Bayes Methods MT25

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Lecture 7: Markov Chain Monte Carlo Methods II

The Gibbs sampler

Random scan Gibbs takes $\xi_i = 1/p$ and $q_i(\theta_i'|\theta) = p(\theta_i'|\theta_{-i})$.

$$p(\theta_i'|\theta_{-i}) = \frac{p(\theta')}{p(\theta_{-i})}$$

is the conditional density, with $\theta_j' = \theta_j,\ j \neq i$.

MH-MCMC with Gibbs proposals is as follows:
Let $X_t = \theta$. Then X_{t+1} is determined in the following way.

(1). Simulate $i \sim U\{1, ..., p\}$ and $\theta_i' \sim p(\cdot|\theta_{-i})$. Set $\theta'_{-i} = \theta_{-i}$.

(2). Set $X_{t+1} = \theta'$.

The acceptance probability is one!* We still check irreducibility.

*Exercise: write down $\alpha(\theta'|\theta)$ for this case and show it equals one.

The *sequential-scan Gibbs sampler* visits each variable θ_i in turn from $i = 1...p$ - also stationary for $p(\theta)$.

Example: Bivariate density $p(\theta_1, \theta_2)$ and $X_t = (\theta_1, \theta_2)$.

Algorithm:

(1) Simulate $\theta'_1 \sim p(\theta'_1|\theta_2)$ then $\theta'_2 \sim p(\theta'_2|\theta'_1)$.

(2) Set $X_{t+1} = (\theta'_1, \theta'_2)$.

Claim: if $X_t \sim p(\cdot)$ then after these two steps we have a new correlated sample $X_{t+1} \sim p(\cdot)$, so the process is stationary. As always irreducibility must be checked.

Suppose $X_{t+1} \sim P(\cdot)$. Show P is p:

$$\begin{aligned}
P(\theta_1', \theta_2') &= \int p(\theta_1, \theta_2) p(\theta_1'|\theta_2) p(\theta_2'|\theta_1') d\theta_1 d\theta_2 \\
&= \int p(\theta_1, \theta_2) \frac{p(\theta_1', \theta_2)}{p(\theta_2)} \frac{p(\theta_1', \theta_2')}{p(\theta_1')} d\theta_1 d\theta_2 \\
&= \int p(\theta_1|\theta_2) p(\theta_2|\theta_1') p(\theta_1', \theta_2') d\theta_1 d\theta_2 \\
&= p(\theta_1', \theta_2') \quad \text{as } p(\theta_1|\theta_2) \text{ and } p(\theta_2|\theta_1') \text{ are normalised.}
\end{aligned}$$

If $\theta^{(0)} = (\theta_1, \theta_2)$, then $\theta^{(1)} = (\theta_1', \theta_2')$ and we can iterate to simulate $\theta^{(2)}, \theta^{(3)}\ldots$. The same proof works for $p > 2$ variables.

Data Augmentation (DA)

DA is convenient when the likelihood on the full data is much simpler than the likelihood on the observed data.

Suppose the observation model is

$$z \sim p(z|\theta), \qquad y \sim p(y|z,\theta)$$

and we observe y. The posterior $\pi(\theta|y)$ is awkward as the likelihood function is an integral,

$$\pi(\theta|y) \propto \pi(\theta) \int p(y|z,\theta)p(z|\theta)dz$$

In data augmentation we target the joint posterior $p(\theta, z|y)$. The "missing data" z are just another parameter. The posterior is

$$\pi(\theta, z|y) \propto p(y|z,\theta)p(z|\theta)\pi(\theta)$$

MCMC targeting $\pi(\theta, z|y)$ gives $(\theta^{(t)}, z^{(t)})$, $t = 1, \ldots, T$ with $\theta^{(t)} \sim \pi(\cdot|y)$ at large t.

A Gibbs sampler for Probit regression - DA simplifies sampling.

Probit regression: data $y_i, i = 1, ..., n$, covariates $x_i = (x_{i,1}, ..., x_{i,p})$, parameters $\theta = (\theta_1, \theta_2, ..., \theta_p)$, linear predictor $\eta_i = \sum_j \theta_j x_{i,j}$, observation model

$$y_i \sim Bern(\Phi(\eta_i))$$

with $E(Y_i) = \Phi(\eta_i)$ the cdf of $N(0, 1)$.

If the prior for θ is $\pi(\theta)$ then the posterior for $\theta|y$ is

$$\pi(\theta|y) \propto \pi(\theta) \prod_i \Phi(\eta_i)^{y_i} (1 - \Phi(\eta_i))^{1-y_i}$$

with $\eta_i = \eta_i(\theta), i = 1, ..., n$. We cant calculate conditionals $\pi(\theta_j|\theta_{-j}, y)$ as θ appears inside Φ so we cant Gibbs sample.

There is another way to represent this model. If (independently)

$$(z_i|\theta) \sim N(\eta_i, 1), i = 1, ..., n$$

and we set $y_i = 1$ if $z_i > 0$ and $y_i = 0$ if $z_i \leq 0$ then*

$$\Pr(y_i = 1|\theta) = \int_{z_i > 0} N(z_i; \eta_i, 1) dz_i \;=\; \Phi(\eta_i).$$

The joint posterior augmented with z is

$$\pi(\theta, z|y) \propto \pi(\theta)\pi(z|\theta) \prod_i \mathbb{I}_{y_i = \mathbb{I}_{z_i > 0}}.$$

The marginal for θ is just $\pi(\theta|y)$, since

$$\pi(\theta|y) \propto \pi(\theta) \prod_i \int \pi(z_i|\theta) \mathbb{I}_{y_i = \mathbb{I}_{z_i > 0}} dz_i \propto \pi(\theta) p(y|\theta)$$

$^*z = \eta + W$ with $W \sim N(0, 1)$ so $\Pr(z > 0) = \Pr(W > -\eta) = \Phi(\eta)$.

Consider Bayesian inference with a normal prior $\theta \sim N(0, \Sigma)$.

$$\pi(\theta, z|y) \propto \pi(z|\theta)\pi(\theta) \prod_i \mathbb{I}_{y_i = \mathbb{I}_{z_i > 0}}.$$

$\pi(\theta|y, z) \propto \pi(z|\theta)\pi(\theta)$ is jointly normal† (conjugate prior) and

$$\pi(z_i|y, \theta) \propto N(z_i; \eta_i, 1)\mathbb{I}_{y_i = \mathbb{I}_{z_i > 0}}$$

Here is the Gibbs sampler. Suppose $X_t = (\theta, z)$ and $\eta = \eta(\theta)$.

(1). For $i = 1, ..., n$, simulate $z_i' \sim N(\eta_i, 1)\mathbb{I}_{y_i = \mathbb{I}_{z_i' > 0}}$.

(2). Simulate $\theta' \sim \pi(\theta|z')$ (multivariate normal, no y given z').

(3) Set $X_{t+1} = (\theta', z')$.

†See PS2 and R code example for this lecture.

Convergence and mixing

We want to estimate $E_p(f(X))$ using our MCMC samples

$$X_0, X_1, X_2, ..., X_n$$

targeting $p(x)$ and calculate the estimate

$$\bar{f}_n = n^{-1} \sum_t f(X_t).$$

The ergodic theorem tells us this estimate converges almost surely to $E_p(f(X))$.

How large should we take n? There are two issues, bias and variance, respectively "convergence" and "mixing".

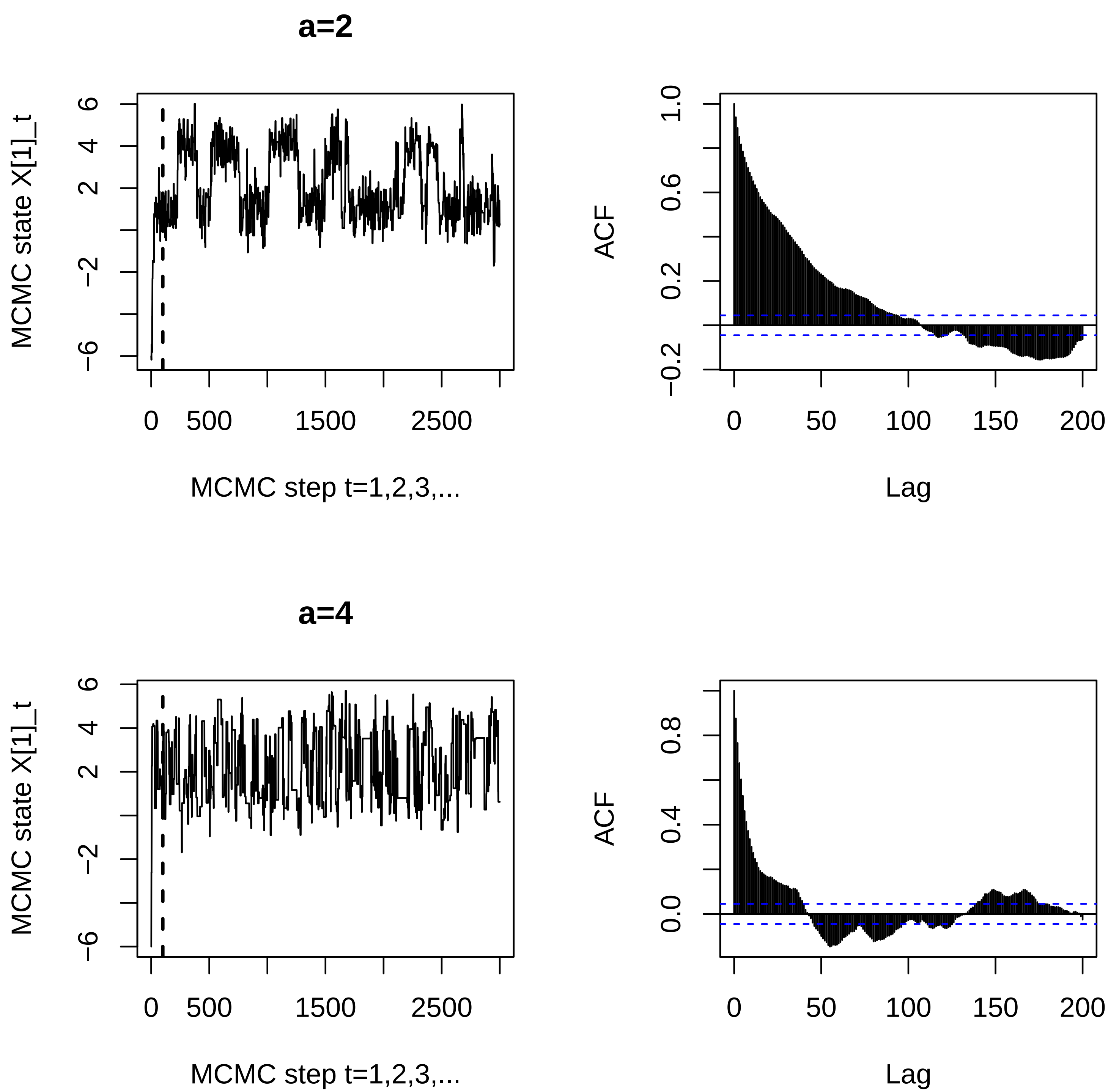
Convergence (bias)
First samples biased by initialization, so $p^{(t)} \neq p$ at small t.

However $p^{(t)} \to p$ so exists T such that $p^{(t)} \simeq p$ for $t \geq T$. Drop "burn-in" $X_1, ..., X_{T-1}$ to reduce initialization bias.

If $n \gg T$ then initialisation-bias in $\bar{f}_n$ is slight.

Mixing (variance)
Second, suppose $p^{(0)}(x) = p(x)$, so we start the chain in equilibrium. The variance, var$(\bar{f}_n)$, of $\bar{f}_n$ will get smaller as n increases. We should choose n large enough to ensure var$(\bar{f}_n)$ is small enough so that $\bar{f}_n$ has useful precision.

The following figures show output from two MCMC runs for the normal mixture with jump size $a = 2, 4$.


a=2
MCMC state X[1]_t
MCMC step t=1,2,3,...
ACF
Lag
a=4
MCMC state X[1]_t
MCMC step t=1,2,3,...
ACF
Lag

MCMC variance in equilibrium* - mixing analysis

$X_0, X_1, X_2, ...$ are correlated so generally $\text{var}(\bar{f}_n) \neq \text{var}(f(X))/n$.

Suppose the chain has reached equilibrium. Correlation at lag s

$$\rho_s^{(f)} = \frac{\text{cov}(f(X_t), f(X_{t+s}))}{\text{var}(f(X_t))}$$

(so $\rho_0 = 1$). Let $\sigma_f^2 = \text{var}(f(X_t))$, $X_t \sim p$. This doesn't depend on t as the chain is stationary.

Express $\text{var}(\bar{f}_n)$ in terms of $\rho_s^{(f)}$. This gives insight and leads to an estimator for $\text{var}(\bar{f}_n)$, since we can estimate $\rho_s^{(f)}$.

*see CJ Geyer "Practical Markov Chain Monte Carlo" Statistical Science 1992 and Sokal Cargèse summer school lecture notes "Monte Carlo Methods in Statistical Mechanics" 1996 for much useful insight.

$$
\begin{aligned}
\mathsf{var}(\overline{f}) &= n^{-2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathsf{cov}(f(X_i), f(X_j)) \\
&= \sigma_f^2 n^{-2}\sum_{i=1}^{n}\sum_{j=1}^{n}\rho_{|i-j|} \\
&= \sigma_f^2 n^{-1}\left[1 + 2\sum_{s=1}^{n-1}\left(1-\frac{s}{n}\right)\rho_s\right]
\end{aligned}
$$

so by dominated convergence

$$
\begin{aligned}
\lim_{n\to\infty} n\mathsf{var}(\overline{f}) &= \sigma_f^2\left[1 + 2\sum_{s=1}^{\infty}\rho_s\right] \\
&= \sigma_f^2\tau_f,
\end{aligned}
$$

Here τ_f is the Integrated Autocorrelation Time (IACT) and ESS$= n/\tau_f$ is the Effective Sample Size - the number of independent samples giving the same precision for $\overline{f}_n$ as the n correlated samples we have.

We can estimate* $\gamma_s = \mathrm{cov}(f(X_i), f(X_{i+s}))$ using

$$\hat{\gamma}_s = \frac{1}{n} \sum_{i=1}^{n-s} (f(X_i) - \hat{f})(f(X_{i+s}) - \hat{f}),$$

and $\gamma_0 = \mathrm{var}(f(X_i))$ (as usual) from the sample output, and compute $\hat{\rho}_s = \hat{\gamma}_s / \hat{\gamma}_0$.

We get an estimate of τ_f,

$$\hat{\tau}_f = 1 + 2 \sum_{s=1}^{M} \hat{\rho}_s,$$

with M a cut-off on the sum. $\hat{\rho}_s$ goes to zero with s and is dominated by noise at large s; terms at large s just add noise. We truncate the sum over s, at $s = M$. We get $\hat{\tau}_f$ consistent if M is chosen according to suitable criteria.

*Priestly 1981 "Spectral Analysis and Timeseries" Academic London, pp323

MCMC convergence

There is no simple generic sufficient condition we can test for convergence. Here some checks we should run to detect poor mixing and identify a burn-in and run length.

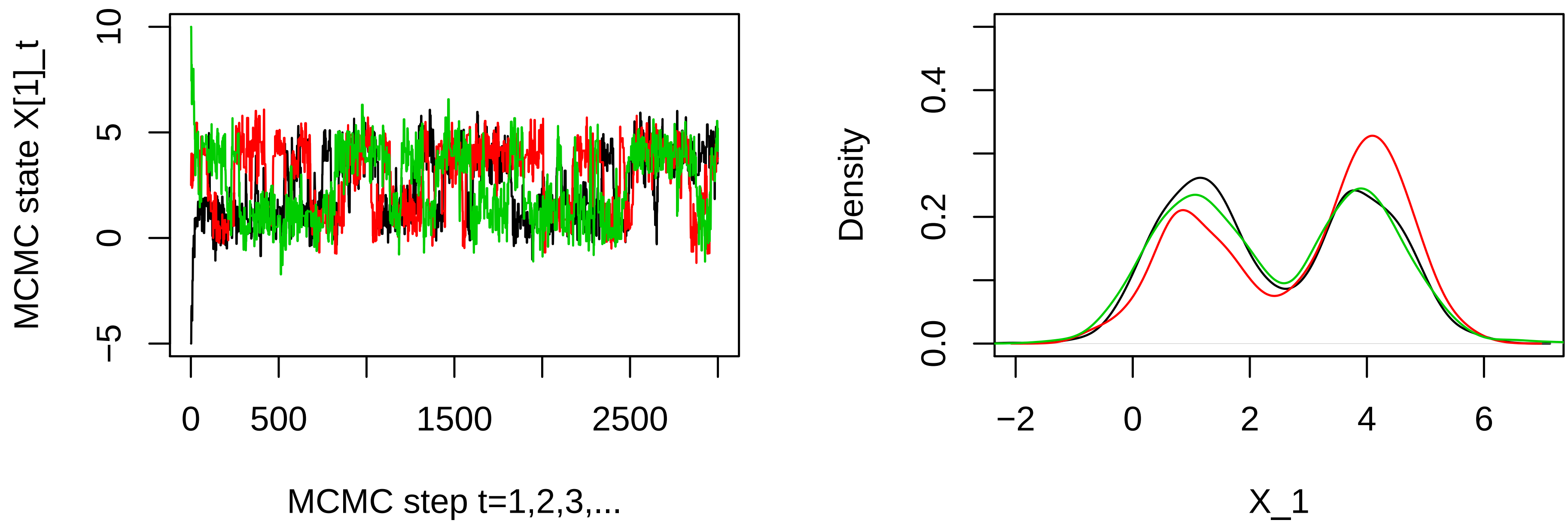
[1] Make multiple runs from different start states and check marginal distributions agree.

[2] Plot the autocorrelation function. Check that it falls off to vary around zero. Calculate the ESS and check it is big enough.*

[3] Plot MCMC traces of the variables and key functions. The chain should be stationary after burn-in.

*Large enough to give you the $\bar{f}_n$ precision you want.

Here is an example of some of the plots I would use for convergence checking on the normal mixture MCMC sampler. In this example the convergence is quite poor: the three histograms differ by more than the precision I would typically be aiming for.



See associated R-file for further examples.